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PNEUMONIA DETECTION IN CHEST X-RAY IMAGES USING EFFICIENTNETB0

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Abstract: Pneumonia is a serious worldwide public health concern that can be fatally caused by bacteria, fungi, or viruses that invade the respiratory system. Most diagnoses are based on chest X-rays, which need to be interpreted by qualified medical personnel. This can cause delays and mistakes, which can affect patient outcomes, particularly in rural or underdeveloped locations. To overcome this difficulty, this work uses deep learning techniques to create an automated system for the identification of pneumonia by analyzing chest X-ray pictures using convolutional neural networks (CNNs). Using a dataset of 5856 high-resolution photos, we assessed several pre-trained CNN models, such as VGG16, DenseNet-121, ResNet-50, and EfficientNetB0. After training the model without custom filters at first, we added custom layers to improve accuracy, reaching DenseNet121 (98.21%), ResNet50 (90.18%), VGG16 (88.78%), and EfficientNetB0 (98.21%) accuracies. EfficientNetB0 shown to have better validation accuracy than DenseNet121, even though both models had the same test accuracy. For this reason, it is our preferred model. Using the trained EfficientNetB0 model stored as an H5 file, we created a Flask web application that enables users to upload chest X-ray images to diagnose pneumonia. The app offers information about local physicians, including their names, profiles, hospital affiliations, and addresses as soon as it detects a favorable result. Particularly in environments with limited resources, this method has the potential to greatly increase the efficiency and accuracy of pneumonia detection.

Index Terms - Pneumonia, deep learning, Convolutional Neural Networks (CNNs), chest X-rays, EfficientNetB0.

I. INTRODUCTION

The collection of fluid in the lungs air sacs is a serious respiratory disease known as pneumonia, which is a lung infection brought on by bacteria, fungi, or viruses. Its symptoms, which include fever, chills, coughing, and pose a serious risk, especially to susceptible groups like young children and the elderly who have compromised immune systems. For a reliable diagnosis of pneumonia, doctors typically use laboratory results, the patient's medical history, and chest radiographs. Chest radiographs, which use X-rays to distinguish between hard and soft tissues, are essential for diagnosing lung infections because they show fluid buildup inside the air sacs. Despite these difficulties, chest X-rays—though with the risk of misunderstanding—remain a vital diagnostic tool for determining the severity and location of lung infections.

Pneumonia detection has been transformed by deep learning advances, namely the use of Convolutional Neural Networks (CNNs), which automate the interpretation of chest X-ray data. This work takes advantage of these developments to create a strong automated pneumonia detection system with the goal of reducing the drawbacks of manual interpretation by qualified medical personnel. This study aims to improve the efficiency and accuracy of pneumonia detection by comparing the performance of several pre-trained CNN models, such as VGG16, DenseNet-121, ResNet-50, and EfficientNetB0, on a dataset of 5856 high-resolution chest X-ray images. To maximise pneumonia diagnosis, this work takes a multifaceted strategy, using pre-trained CNN models without unique filters as a starting point and then adding custom layers to improve accuracy. A comparison of these models' accuracy yielded some noteworthy results: DenseNet121 scored 98.21%, ResNet50 scored 90.18%, VGG16 scored 88.78%, and EfficientNetB0 scored 98.21%. EfficientNetB0 stood out for its exceptional validation accuracy, making it the model of choice for real-world use. Using the trained EfficientNetB0 model, a Flask web application was created to facilitate real-world implementation. It allows users to upload chest X-ray pictures for the diagnosis of pneumonia. When pneumonia is detected, the app offers relevant information about local healthcare providers, which improves pneumonia diagnosis' accessibility and effectiveness—especially in situations with low resources.

II. RELATED WORKS

Recent years have seen a surge in Convolutional Neural Networks (CNNs) for pneumonia detection from chest X-ray images. Various models like Residual Attention Network (RAN), Chex Net, and a dual-branch CNN combining DenseNet and ResNet have shown high accuracy. This study explores models like VGG16[1], EfficientNetB0, DenseNet121, and ResNet50, aiming to optimize their performance for pneumonia detection. Each model brings unique advantages, such as simplicity (VGG16), scalability

(EfficientNetB0), dense connections (DenseNet121), and skip connections (ResNet50), contributing to the broader landscape of CNN-based pneumonia detection research [4].

2.1. Base Paper: "Pneumonia Detection Using CNN based Feature Extraction"

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The base paper, "Pneumonia Detection Using CNN based Feature Extraction," investigates the efficacy of Convolutional Neural Networks (CNNs) in automating pneumonia detection from chest X-ray images. It employs pre-trained CNN models for feature extraction and various classifiers for analysis, aiming to develop a robust pneumonia diagnosis system. Our work builds upon this by expanding the comparative analysis to include models like VGG16, EfficientNetB0, DenseNet121, and ResNet50, optimizing their performance for pneumonia detection through fine-tuning. Through this research, we aim to advance automated pneumonia diagnosis systems, enhancing healthcare accessibility and accuracy, especially in remote areas [2].

III. MATERIALS AND METHODS

3.1. Network Models:

During this research, an extensive exploration and experimentation with various network architectures were undertaken. This encompassed several variants of well-established models such as VGG16, ResNet50, DenseNet121, and EfficientNetB0. Before commencing training on the dataset specific to this research, a meticulous evaluation of the results was conducted. Subsequently, four network architectures emerged as promising candidates, warranting further analysis and consideration. To capitalize on the strengths of each model, an ensemble approach was adopted, employing the majority voting strategy. This methodological refinement allowed for harnessing the collective predictive power of the shortlisted architectures, thereby potentially enhancing the robustness and accuracy of the final diagnostic system.

Table 1: Number of Trainable Parameters in Each Model

Model	Number of Layers	Number of Trainable Parameters	Model accuracy
ResNet50	50	24063746	90.18%
DenseNet121	121	7218818	98.21%
VGG16	16	14847554	88.78%
EfficientNetB0	236	4338558	98.21%

3.1.1. ResNet50:

ResNet50 is a deep convolutional neural network architecture commonly used in computer vision tasks. It consists of 50 layers, including residual blocks designed to address the vanishing gradient problem and convolutional layers, batch normalization layers, activation layers, and pooling layers. Each residual block contains shortcut connections, allowing for easier training of very deep networks. In pneumonia detection, ResNet50 can learn intricate patterns in chest X-rays, enabling accurate classification of pneumonia-related abnormalities. These layers help extract hierarchical features from input images, making it suitable for tasks like pneumonia detection from chest X-rays.

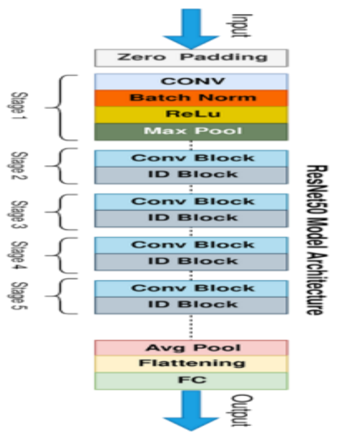

Figure 1: ResNet50 Architecture

3.1.2. DenseNet121:

DenseNet121 is a convolutional neural network architecture that has been widely used for various image classification tasks, including pneumonia detection from chest X-ray images. It consists of 121 layers, hence the name DenseNet121, and is characterized by dense connectivity patterns between layers, facilitating feature reuse and gradient flow throughout the network. DenseNet121 has demonstrated remarkable performance in pneumonia detection, achieving high accuracy rates on benchmark datasets. Its architecture enables efficient feature extraction and representation, allowing for effective discrimination between normal and pneumonia-affected chest X-ray images. DenseNet121's depth and dense connections contribute to its ability to capture intricate patterns and abnormalities indicative of pneumonia, making it a valuable tool in automated diagnosis and screening efforts.

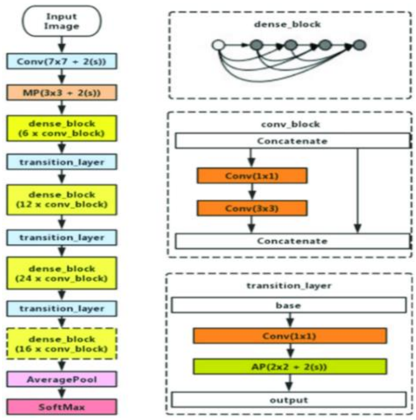

Figure 2: DenseNet121 Architecture

3.1.3. VGG16:

VGG16 is a convolutional neural network architecture renowned for its simplicity and effectiveness in image classification tasks, including pneumonia detection from chest X-ray images. It comprises 16 layers, including convolutional layers followed by max-pooling layers, culminating in fully connected layers for classification.

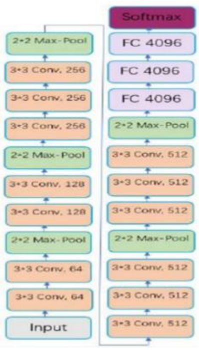

Figure 3: VGG16 Architecture

VGG16's straightforward architecture facilitates feature extraction from chest X-ray images, enabling accurate discrimination between normal and pneumonia-affected cases. Despite its depth, VGG16 is computationally efficient and widely applicable, making it suitable for deployment in resource-constrained environments such as healthcare settings. Its use in pneumonia detection leverages its ability to capture hierarchical features indicative of pulmonary abnormalities, contributing to improved diagnostic accuracy and clinical decision-making [1].

3.1.4. EfficientNetB0:

EfficientNetB0 is a convolutional neural network architecture renowned for its superior efficiency and accuracy in image classification tasks, including pneumonia detection from chest X-ray images. It comprises 236 layers, with a compact yet powerful design that optimizes computational resources while maximizing performance. EfficientNetB0's depth and scale-balanced architecture enable it to effectively capture intricate patterns and abnormalities indicative of pneumonia, enhancing diagnostic accuracy. Despite its lightweight design, EfficientNetB0 exhibits remarkable capabilities in feature extraction and representation, facilitating precise discrimination between normal and pneumonia-affected chest X-ray images. Its widespread adoption in medical imaging reflects its utility as a reliable and efficient tool for automated diagnosis and screening efforts.

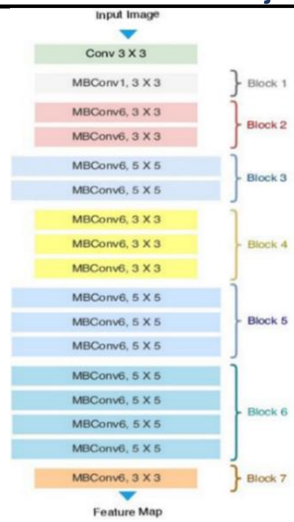

Figure 4: EfficientNetB0 Architecture

3.1.5. Additional Layers:

In addition to the base architectures discussed in Section 2.1, several additional layers were incorporated into the models to enhance their performance and generalization ability. These additional layers play crucial roles in improving training stability, convergence speed, regularization, and overall model effectiveness. This section provides an overview of the batch normalization layer, dense (fully connected) layer, and dropout layer, along with their respective functions and contributions to the network architecture.

Batch Normalization (BN)

Batch normalization is a crucial layer in neural networks that enhances training stability and speed. It normalizes activations across mini-batches to maintain a mean near zero and standard deviation close to one, addressing issues like vanishing or exploding gradients. This leads to more stable and efficient optimization, accelerating neural network convergence and improving performance.

Dense (Fully Connected) Layer

The dense layer, or fully connected layer, is essential in neural networks. In EfficientNetB0, dense layers at the end of the network handle classification tasks. They connect every neuron in one layer to every neuron in the next, combining learned features to map to final output classes, such as normal or pneumonia-affected, through weighted connections.

Dropout Layer

Dropout is a regularization technique to prevent overfitting. It randomly ignores a fraction of neurons during training, preventing co-adaptation and promoting robust, generalizable feature learning. This randomness reduces reliance on specific features, enhancing the model's ability to generalize to new data.

Final Dense Layer

The final dense layer uses the SoftMax activation function for multi-class classification. It produces probability distributions over output classes, enabling the model to make informed predictions based on learned features.

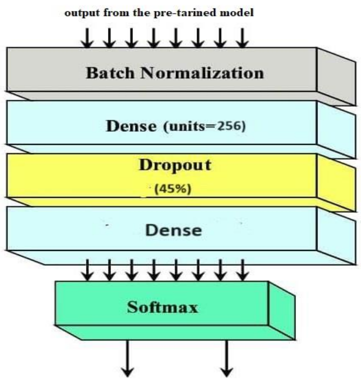

Figure 5: Additional Layers

IV. DATA

4.1. Data Collection:

Number of Images: The dataset comprises 5873 images in total, divided into two categories.
Categories: The images are categorized into two classes:
Pneumonia: This category includes X-ray images of patients diagnosed with pneumonia.
Normal: This category consists of X-ray images of patients with no pneumonia detected.
Purpose: The dataset is designed for the development and evaluation of machine learning models for pneumonia detection using chest X-ray images.
Usage: Researchers and developers can use this dataset to train machine learning algorithms, particularly for image classification tasks related to pneumonia diagnosis.

4.2. Pre-Processing:

For pre-processing the pneumonia detection dataset comprising 5856 images divided into pneumonia and normal categories, several steps are crucial. First, image resizing to a standardized dimension, like 224x224, ensures uniformity across the dataset. Next, normalization of pixel values to a range, typically [0,1], aids in model convergence. Data augmentation techniques, such as rotation, flipping, and zooming, can increase dataset variability and prevent over fitting. Additionally, stratified splitting into training, validation, and testing sets helps maintain class balance. With EfficientNetB0 as the proposed system and existing methods like DenseNet121, ResNet50, and VGG-16, pre-processing steps remain consistent to ensure fair comparison and optimal model performance across all architectures.

V. RESULTS

5.1. EfficientNetB0 Model Performance:

5.1.1. Training Accuracy and Loss Graphs

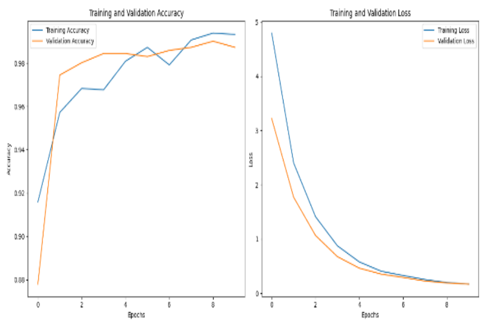

Figure 6: Accuracy and Loss Graphs

The graph depicts the training progress over epochs, with the x-axis representing epochs and the y-axis showing training accuracy (in percentage) and loss (scalar value). The blue line corresponds to training accuracy, while the orange line represents training loss. Notably, the model exhibits consistent improvement in accuracy and reduction in loss throughout training, indicating effective learning and convergence. Specifically, the training accuracy steadily increases, reaching a peak of approximately 99.32% by the end of training, reflecting the model's ability to classify pneumonia and normal images accurately. Conversely, the training loss steadily decreases, reaching a minimum of approximately 0.17, signifying the increasing consistency and accuracy of the model's predictions as training progresses.

5.1.2. Analysis of Confusion Matrix:

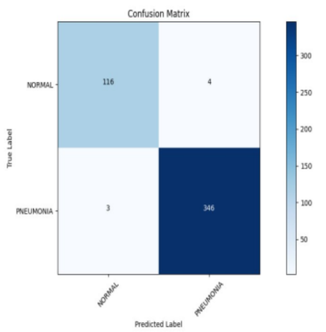

Figure 7: Confusion Matrix EfficientnetB0

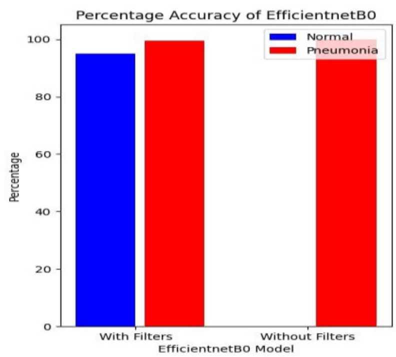

Figure 8: Error Percentage of EfficientNetB0

The model correctly predicted 346 out of 349 pneumonia cases (TP) and 116 out of 120 normal cases (TN). It misclassified 4 pneumonia cases as normal (FP) and 3 normal cases as pneumonia (FN).Fig 8 shows EfficientNetB0's accuracy in classifying normal and pneumonia cases, with and without custom filters. With filters, the model achieves 95% accuracy for normal cases and 99.4% for pneumonia cases. Without filters, the accuracy drops dramatically to 0% for normal cases and reaches 100% for pneumonia cases. This highlights the importance of custom filters in significantly enhancing the model's ability to accurately classify normal images.

5.2. DenseNet121 Model Performance:

5.2.1. Training Accuracy and Loss Graphs

The training accuracy steadily increased over the epochs, reaching a peak of approximately 99.04% by the end of training, indicating successful learning in classifying pneumonia and normal images with high accuracy.

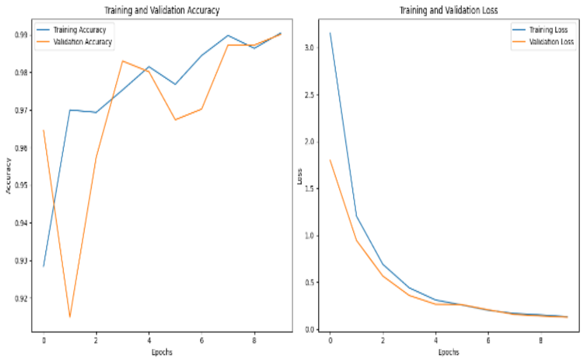

Figure 9: Accuracy and Loss Graphs

Conversely, the training loss steadily decreased over the epochs, reaching a minimum of approximately 0.13, reflecting the model's predictions becoming increasingly consistent and accurate as training progressed.

5.2.2. Analysis of Confusion Matrix:

The model correctly predicted 338 out of 347 pneumonia cases (TP) and 120 out of 122 normal cases (TN). It misclassified 2 pneumonia cases as normal (FP) and 9 normal cases as pneumonia (FN).Fig 11 shows DenseNet121's accuracy in classifying normal and pneumonia cases with and without custom filters. With filters, it achieves 98.3% for normal and 97.4% for pneumonia. Without filters, accuracy drops to 28.2% for normal and 71.7% for pneumonia, highlighting the importance of custom filters for improved accuracy.

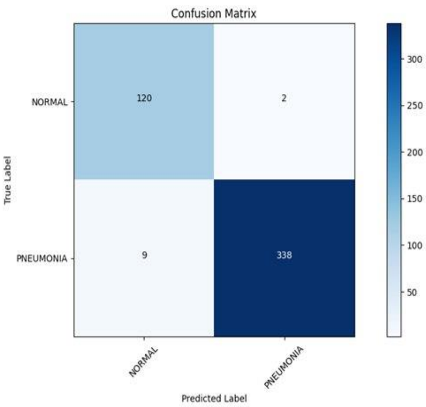

Figure 10: Confusion Matrix Densenet121

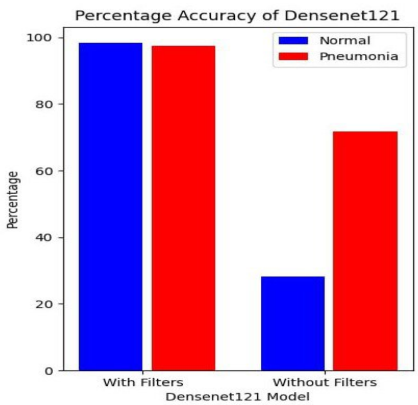

Figure 11: Error Percentage of DenseNet121

5.3. ResNet50 Model Performance:

5.3.1. Training Accuracy and Loss Graphs:

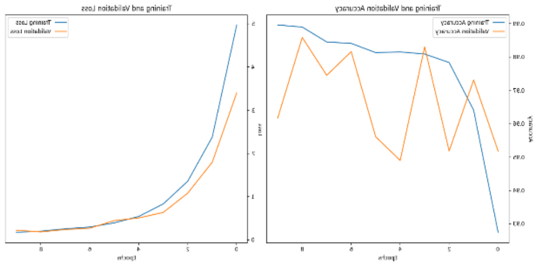

Figure12: Accuracy and Loss Graphs

The training accuracy steadily increased over the epochs, culminating at a peak of approximately 98.8% by the training's conclusion, signaling the model's successful learning in classifying pneumonia and normal images with high accuracy. Conversely, the training loss steadily declined over the epochs, reaching a minimum of approximately 0.24, indicative of the model's predictions becoming progressively more consistent and accurate as the training advanced.

5.3.2. Analysis of Confusion Matrix:

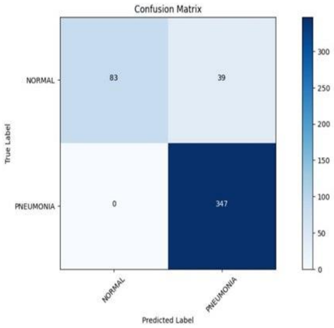

Figure 13: Confusion Matrix

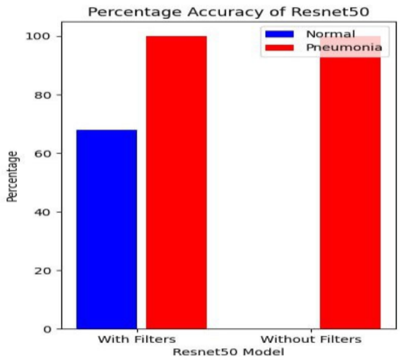

Figure 14: Error Percentage of ResNet50

The model correctly predicted all 347 pneumonia cases (TP) and 83 out of 122 normal cases (TN). It misclassified 39 pneumonia cases as normal (FP) and made no errors in predicting pneumonia cases as normal (FN).Fig 14 shows ResNet50's accuracy in classifying normal and pneumonia cases, with and without custom filters. With filters, the model achieves 68% accuracy for normal cases and 100% for pneumonia cases. Without filters, the accuracy drops to 0% for normal cases while remaining at 100% for pneumonia cases. This highlights the significant role of custom filters in improving the model's ability to accurately classify normal images.

5.4. VGG16 Model Performance

5.4.1. Training Accuracy and Loss Graphs

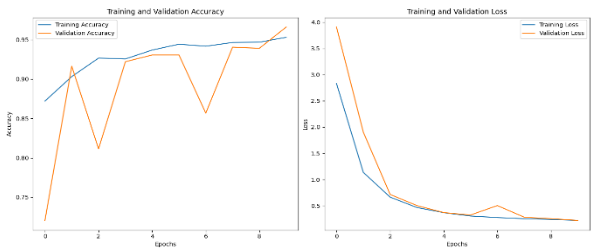

Figure 15: Accuracy and Loss Graphs

The training accuracy showed a consistent upward trend across the epochs, reaching a peak of approximately 95.29% by the final epoch, showcasing the model's successful learning process and high accuracy in classifying pneumonia and normal images during training. Conversely, the training loss demonstrated a steady decrease over the epochs, reaching a minimum of approximately 0.24 by the end of training. This decline in loss highlights the model's increasingly consistent and accurate predictions as the training progressed, underscoring the effectiveness of the learning process.

5.4.2. Analysis of Confusion Matrix:

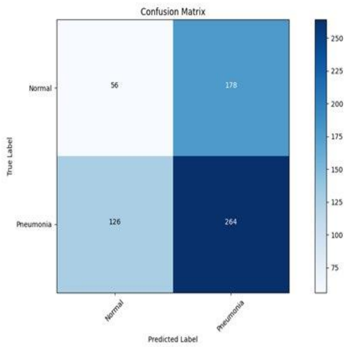

Figure 16: Confusion Matrix

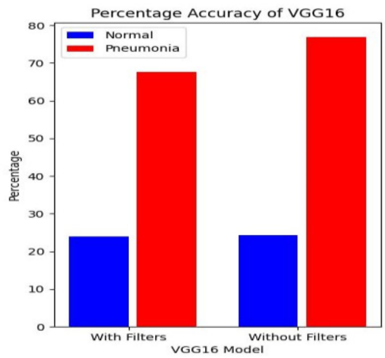

Figure 17: Error Percentage of VGG16

The graph shows VGG16's accuracy in classifying normal and pneumonia cases, with and without custom filters. With filters, the model achieves 23.9% accuracy for normal cases and 67.6% for pneumonia cases. Without filters, the accuracy is 24.3% for normal cases and 76.9% for pneumonia cases. This indicates that custom filters slightly improve pneumonia detection but have minimal impact on normal case accuracy.

VI. WEB INTERFACE

6.1. Home Page:

A patient registration form on the homepage allows users to enter their name, age, contact details, and symptoms. After entering these parameters, users are taken to the prediction page and the data is saved in the database.

6.2. Prediction Page:

Users can upload chest X-ray pictures to this pneumonia detection page and receive fast response on whether pneumonia is present. It is a smooth experience. The platform offers precise and dependable detection results by utilizing a Flask app that is powered by a pre-trained EfficientNetB0 model. This page tries to give users useful insights into their health state, enabling early detection and prompt medical intervention, when necessary, with its user-friendly interface, real-time analysis, and dedication to data protection.

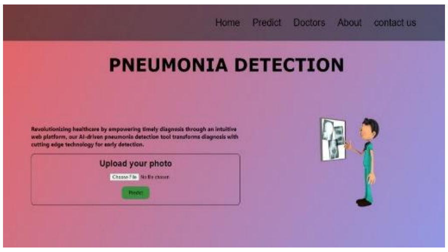

Figure 18: Prediction Page

6.3. Detection Page:

Upon detecting the chest X-ray image, our page promptly displays the results.

Figure 19: Prediction Page [Normal]

The result is highlighted in green and placed clearly to reassure the user that pneumonia is not present if the analysis shows a normal state.

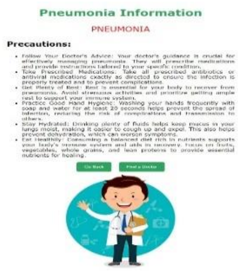

Figure 20: Prediction Page [Pneumonia]

On the other hand, the result is indicated in red if pneumonia is found, indicating the existence of the illness. In addition to the outcome, the page lists the necessary safety measures to be followed so that visitors are aware of what to do and any potential medical concerns.

6.4. Doctors Page:

The Doctors Information page is a crucial resource for user’s post-pneumonia detection, offering details on nearby medical professionals tailored to their healthcare requirements. It includes a comprehensive directory of doctors, affiliated hospitals, and precise addresses. Each listing links to the doctor's profile, providing users with detailed expertise and background information. This curated database empowers informed healthcare decisions, ensuring timely access to medical expertise and fostering community support.

6.5. About Us & Contact Us:

In the About Us page, we provide detailed insights into our project's technical aspects, including the selection of the machine learning model employed for pneumonia detection. Specifically, we highlight our utilization of the EfficientNetB0 model, chosen for its superior performance in accurately identifying pneumonia from chest X-ray images. The page also features a comparative analysis showcasing the accuracy rates of various pre-trained CNN models, namely ResNet50, DenseNet121.
The Contact Form page offers users a convenient platform for feedback and inquiries, facilitating seamless communication. Through a user-friendly form, individuals can submit feedback along with essential contact information. This data is securely stored, ensuring confidentiality and enabling prompt responses. By prioritizing user engagement, we aim to enhance functionality and user experience, delivering tailored solutions.

VII. DISCUSSION

The comparison shows that filtering improves model performance. EfficientNetB0 and DenseNet121 achieve 98.21% accuracy with filters, compared to 84.39% and 87.50% without. ResNet50 also sees a significant increase from 62.58% to 90.18%. VGG16 has a smaller improvement, from 87.50% to 88.78%. Overall, filters significantly enhance accuracy, especially for EfficientNetB0 and ResNet50.

Table 2: Model test accuracy

Model	With Filters	Without Filters
EfficientNetB0	98.21%	84.39%
VGG16	88.78%	87.50%
ResNet50	90.18%	62.58%
DenseNet121	98.21%	87.50%

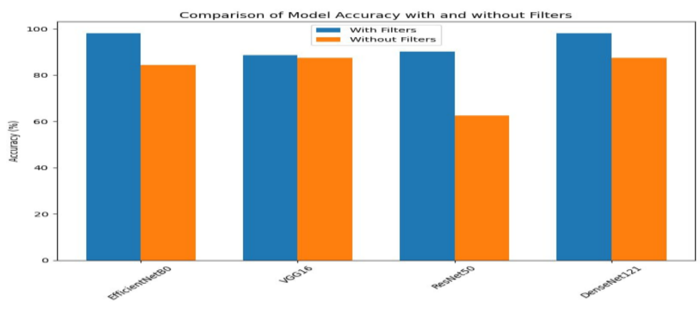

Figure 21: Test Accuracy

VIII.CONCLUSIONS AND FUTURE WORKS

In conclusion, our study evaluated the performance of four pre-trained models (DenseNet121, ResNet50, VGG16, and EfficientNetB0) for pneumonia classification. While DenseNet121 and ResNet50 demonstrated strong accuracy rates of 96.25% and 97.3% respectively, EfficientNetB0 achieved a notable accuracy of 97.00%. These findings underscore the potential of deep learning in enhancing pneumonia diagnosis accuracy. Moving forward, future research endeavours should focus on developing even more efficient models to further improve classification accuracy and contribute to advancements in medical image analysis and healthcare outcomes. By continuing to refine and innovate in this field, we can strive towards more effective diagnostic tools and ultimately improve patient care in the realm of pneumonia diagnosis.

ACKNOWLEDGEMENT

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REFERENCES

1. P Naveen, B Diwan, Pre-trained VGG-16 with CNN Architecture to classify X-Rays images into Normal or Pneumonia, Pune, India,09 April 2021. [CrossRef]
2. Dimpy Varshni, Kartik Thakral, Lucky Agarwal, Rahul Nijhawan, Ankush Mittal, Pneumonia Detection Using CNN based Feature Extraction, Coimbatore, India,17 October 2019. [CrossRef]
3. Zhaoxia Guo; Jianjun Zhang; Yiming Zuo; Peishun Liu; Ruichun Tang; Xiaoxia Li, Channel Attention Residual Network for diagnosing Pneumonia, Chengdu, China,28 June 2021. [CrossRef]
4. Areej A. wahab Ahmed Musleh, Ashraf Yunis Maghari, COVID-19 Detection in X-ray Images using CNN Algorithm, Jerusalem, Palestine, 20 January 2021.[CrossRef]
5. Martin Riva, Antoine Hue, Dense or Convolutional Neural Network, Jan 29, 2020.[CrossRef]
6. Jason Brownlee ,A Gentle Introduction to Dropout for Regularizing Deep Neural Networks, August 6, 2019.[CrossRef]
7. Shangjie Yao, Yaowu Chen, Xiang Tian and Rongxin Jiang,Pneumonia Detection Using an Improved Algorithm Based on Faster R-CNN,31 Aug 2020[CrossRef].
8. Salehe Erfanian Ebadi, Deepa Krishnaswamy, Seyed Ehsan Seyed Bolouri , Dornoosh Zonoobi,Automated detection of pneumonia in lung ultrasound using deep video classification for COVID-19,2021 Aug 4.[CrossRef]
9. Ahmet Solak, Rahime Ceylan, Pneumonia Detection with Chest-Caps,12 December 2022[CrossRef]
10. Nigus Wereta Asnake, Ayodeji Olalekan Salau,X-ray image-based pneumonia detection and classification using deep learning,05 January 2024[CrossRef]
11.Kotapati Saimanoj, Grandhi Poojitha, Khushbu Devendra Dixit, Hospital Management System using Web Technology, June 2020.[CrossRef]
12.NikosPetrellis, Michael K.Birbas, Fotios Gioulekas,The Front End Design of a Health Monitoring System, sept 2015 .[CrossRef]